

# AMENDMENT NO. 5      OCTOBER 1988

## TO

### IS : 2825 - 1969 CODE FOR UNFIRED PRESSURE VESSELS

( Page 9, clause 1.3.1.3, line 4 ) — Add the words 'or gas' after the word 'vapour'.

( Page 9, clause 1.3.1.3 )

- a) Line 4 — Add the words 'or gas' after the word 'vapour'.
- b) Line 5 — Delete the word 'design'.

NOTE — Hydrostatic pressure means pressure due to liquid head only.

[ Page 14, Tables 3.1, 3.2, 3.3 and 3.4 ( see also Amendment No. 2, Sl No. 11 ) ] — Retain the tables with all the existing matter.

[ Page 21, clauses 3.4.6.1(a) and 3.4.6.2(a), line 4 ( see also Amendment No. 2, Sl No. 12 and 13 ) ] — Substitute '3.3.3.1' for '3.3.3.3'.

[ Page 35, clause 3.8.7.1(b), line 2 ] — Substitute 'P2 ( 50 mm nominal bore )' for 'P 1½ size'.

( Page 36, clause 3.9.4, last line ) — Substitute 'P2 ( 50 mm nominal bore )' for 'size P 1½'.

( Page 65, clause 6.2.5, first para ) — Substitute the following for the existing para:

'Any person who wishes to qualify for welder's performance test under this code, shall not be below the age of 18 years and shall have adequate experience or training.'

( Page 71, clause 6.4.10, last sentence ) — Substitute the following for the existing sentence:

'If the forming or bending operation takes place in hot condition, no subsequent heat treatment is required, provided during the last operation, it is uniformly heated to a temperature within the normalizing range.'

[ Page 71, clause 6.4.11, first sentence ( see also Amendment No. 3, Sl No. 6 ) ] — Substitute the following for the existing sentence:

'6.4.11 Plates Welded Prior to Forming — Seams in plates may be welded prior to forming provided they are examined radiographically throughout the entire length after forming. In the case of class 1 and 2 vessels, the seams shall also be inspected by magnetic crack detection or dye-penetrants.'

( Page 71, clause 6.5.1, last sentence ) — Substitute the following matter for the existing sentence:

'Tack welds shall either be removed completely when they have served their purpose or their stopping and starting ends shall be properly prepared by grinding or other suitable means so that they may be satisfactorily incorporated into the final weld. Tack welds shall be made by qualified procedures and welders, and shall be examined visually for defects and, if found to be defective, shall be removed.'

[ Page 75, clause 6.12.2(a) ] — Substitute the following for the existing matter:

'a) intended for containing lethal\* material;'

[ Page 75, clause 6.12.2(g) ] — Add the following new items after 6.12.2(g).

'h) intended for transport of flammable or toxic material; and

j) when required by the statutory authority.'

( Page 76, Table 6.3 ):

a) Material Group 2a — Substitute 'Sum of alloying elements, that is, Cr, Mo and V 0.80 Max' for 'Residual or other elements 0.80 Max'.

b) Renumber material group '2a' as '2'.

c) Renumber material group '2b' as '3'.

d) Delete the present material group '3'.

[ Page 77, clause 6.12.3.1(b), line 6 ] — Substitute '2.5√rt' for '5√rt'.

( Page 96, clause 8.4.2.2, para 3, first sentence ) — Substitute the following for the existing sentence:

'The vessel shall be maintained at the specified test pressure for a sufficient length of time to stabilize the pressure but in no case less than 10 minutes. A thorough examination of the vessels shall be carried out after lowering the pressure to 80 percent of the test pressure.'

( Page 96, clause 8.4.2.2, para 3, second and third sentences ) — Delete.

( Page 96, clause 8.5.1 ) — Substitute the following for the existing clause:

'8.5.1 Two production test plates representing approximately 15 metres of weld or five shell courses, whichever is higher, shall be provided with longitudinal seams for every weld procedure applicable to the vessels or series of vessels made to the same drawing/specification. Vessels having more than 5 shell courses shall be provided with an additional production test plate representing the next 15 metres of welding or part thereof. No test plate need be provided for circumferential seams except in cases where a vessel has circumferential seams only or the welding process, procedure or technique is substantially different, in which case two test plates are to be provided.'

8.5.1(a) For low temperature vessels, in addition to the test coupon plates mentioned in 8.5.1, each other shell course shall also be provided with a coupon plate sufficiently long for testing two sets of impact test ( one for weld metal and one for heat affected zone ). One set of tests shall be conducted to cover 5 seams.

\*See 1.3.1.1(a) regarding clarification of the word 'lethal'.

**8.5.1(b)** For vessels other than covered under **8.5.1 (a)**, any one test plate shall be selected by the inspector for all tests described in figures under **8.5.1.2** of the code except for all weld metal tensile test.'

[ Page 99, Fig. 8.5(A) ] — Delete.

[ Page 99, Fig. 8.5(B) ] — Delete the letter '(B)' from the title of bottom figure and substitute 'Fig. 8.5' for 'Fig. 8.5(B)' wherever occurs in this code.

( Page 102, clause **8.6.2** ) — Substitute the following for the existing clause:

**8.6.2** The minimum average impact strength value for the impact test piece shown in Fig. 8.5 for V-notch specimen shall be 2.8 kgf.m ( 3.5 kgf m/cm<sup>2</sup>). The minimum impact strength value for any test piece shall be 2.1 kgf.m ( 2.6 kgf.m/cm<sup>2</sup>).

NOTE — The value is equivalent to 2.8 kgf.m for a 10 × 10 mm test piece.'

( Page 103, clause **8.6.8**, line 4 ) — Substitute 'retests' for 'new tests'.

( Page 104, clause **8.7.2.1** )—Add the following at the end of the clause:

'[ see also **8.7.10.3(b)** ]'

( Page 104, clause **8.7.3.3** ) — Substitute the following for the existing clause:

**8.7.3.3** Radiographic examination may be conducted before the final heat treatment. However for chromium molybdenum steels having the chromium content equal to or more than 1.5 percent radiography is recommended after heat treatment.'

( Page 115, Table A.1 )— Delete the reference to IS : 1570-1961 wherever mentioned.

( Page 195, Fig. G.1 ) — Substitute 'Nominal pipe size 50 mm Max' for 'BORE MAX 100 mm NOMINAL'.

[ Page 195, Fig. G.1, foot-note (b) ] — Substitute 'P2 ( 50 mm nominal bore )' for 'size 1½'.

[ Page 195, Fig. G.2, foot-note (b) ] — Substitute 'P2 ( 50 mm nominal bore )' for 'size 1½'.

[ Page 195, Fig. G.3, foot-note (c) ] — Substitute 'P2 ( 50 mm nominal bore )' for 'size 1½'.

( Page 196, Fig. G.4, foot-note (d) ] — Substitute 'P2 ( 50 mm nominal bore )' for 'size 1½'.

[ Amendment No. 4, page 1, clause **P-2.0(c)** ] — Substitute the following for the existing item:

'(c) Dimple jackets ( see Fig. P.6 ); and'

[ Amendment No. 4, page 3, clause **P-2.1(c)**, line 2 ] — Substitute 'only' for 'on'.

( Amendment No. 4, page 3, clause **P-2.2, Heading** ) — Substitute the following for the existing heading:

**'P-2.2 Half Pipe Coil Jackets or Limpet Coil Jackets ( see Fig. P.3, P.4 and P.7 )'**

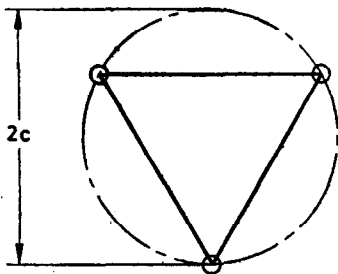
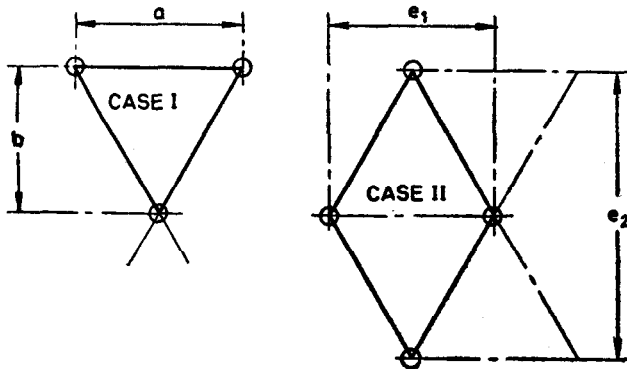
( Amendment No. 4, page 3, clause **P-2.2** )—Add the following new clause after **P-2.2**:

#### **'P-2.2.1 Notation**

| Symbol   | Unit                | Description                                                                                                 |
|----------|---------------------|-------------------------------------------------------------------------------------------------------------|
| $f_{c1}$ | kgf/mm <sup>2</sup> | Hoop stress in the coil due to coil pressure                                                                |
| $f_{c2}$ | kgf/mm <sup>2</sup> | Longitudinal stress in the coil due to coil pressure                                                        |
| $f_{st}$ | kgf/mm <sup>2</sup> | Total hoop stress in the shell                                                                              |
| $f_{s1}$ | kgf/mm <sup>2</sup> | Hoop stress in the shell due to coil pressure                                                               |
| $f_{s2}$ | kgf/mm <sup>2</sup> | Longitudinal stress in the shell due to vessel pressure                                                     |
| $f_{s3}$ | kgf/mm <sup>2</sup> | Longitudinal stress in the shell due to coil pressure                                                       |
| $f_{s4}$ | kgf/mm <sup>2</sup> | Longitudinal stress in the shell due to bending                                                             |
| $a$      | —                   | } Spacing of dimples as per Case 1 ( Uniform spacing of dimples ) ( Fig. P.6 )                              |
| $b$      | —                   |                                                                                                             |
| $c$      | mm                  | Radius of loaded area considered ( Fig. P.6 )                                                               |
| $e_1$    | —                   | } Spacing of dimples as per Case 2 ( Non-uniform spacing of dimples ) ( Fig. P.6 )                          |
| $e_2$    | —                   |                                                                                                             |
| $W$      | kgf                 | Total load uniformly distributed over area of radius 'c' ( Fig. P.6 ) ]                                     |
| $m$      | —                   | Inverse of Poission's ratio                                                                                 |
| $C$      | —                   | Constant concerning the load conditions                                                                     |
| $f_{SL}$ | kgf/mm <sup>2</sup> | Total longitudinal stress in the shell ( in case of half pipe coil type of jackets ) ( see Equation P.12 ). |
| $f_e$    | kgf/mm <sup>2</sup> | Maximum equivalent stress at the half pipe coil to shell junction ( see Equation P.13 )                     |
| $f_{ed}$ | kgf/mm <sup>2</sup> | Maximum stress in the dimple jacket wall'                                                                   |

( Amendment No. 4, page 3, clause P-2.4 ) — Substitute 'l/m' for 'l/m'.

( Amendment No. 4, page 4, Fig. P.6 ) — Delete the two existing figures given on the right hand bottom on this page and add the following new figures.



( Amendment No. 4, page 7, clause P-3.2.4, equation P.13 ) — Substitute the following for the existing equation:

$$f_e^2 = f_{e1}^2 + f_{st}^2 + f_{o1}^2 - f_{s1} \times f_{st} + f_{e1} \times \frac{f_{o1}}{f_{o1} + f_{st} \times f_{e1}} \quad \dots (P.13)$$

( Amendment No. 4, page 7, clause P-3.3 ) — Substitute the following for the existing clause:

### P-3.3 Dimple Jackets

**P-3.3.1 Dimple Plate Thickness** — Dimples can be assumed to act as 'stays' in flat heads and plate thickness is checked below, both for:

- Case 1—Uniform spacing of stays; and
- Case 2—Non-uniform spacing of stays.

**P-3.3.1.1 Case 1—Uniform spacing of stays** ( see Fig. P.6 ).

$$t_{r1} \text{ Min} = C \times \sqrt{\frac{p(a^2 + b^2)}{100 \times f}} \quad \dots (P.14)$$

**P-3.3.1.2 Case 2—Non-uniform spacing of stays** ( see Fig. P.6 )

$$t_{r1} \text{ Min} = C \times \frac{e_1 + e_2}{2} \sqrt{\frac{p}{100 \times f}} \quad \dots (P.15)$$

where  $C = 0.40$  for welded stays.

**P-3.3.2 Checking for Stress and Deflection** ( see Fig. P.6 ) — Considering fixed plate with uniformly distributed load over the entire surface.

### P-3.3.2.1 Maximum stress at the edge,

$$f_{ed} = \frac{3W}{4 \times t_{r1}^2} \quad \dots (P.16)$$

### P-3.3.2.2 Maximum deflection at the centre

$$y = (-) \frac{3 \times W (m^2 - 1) \times c^2}{16\pi \times E \times m^2 \times t_{r1}^2} \quad \dots (P.17)$$

The adopted thickness ( $t_1$ ) shall include corrosion and manufacturing allowances, etc.

**P-3.3.3 Dimple Connecting Welds** — The weld attachment is made by fillet welds around holes, or if the thickness of the plate having hole is 5 mm or less and hole is 25 mm or less in diameter, the holes may be completely filled with weld metal. The allowable load on weld shall equal the product of the thickness of plate having the hole, the circumference of perimeter of hole, the allowable stress value in tension of the material being welded and joint efficiency of 55 per cent. The connecting welds shall be checked for shear and tear failures.

( Pages 79 to 91, Section II, Chapter 7 ) — Substitute the following for the existing matter:

## 7. WELDING QUALIFICATIONS

**7.0 Foreword** — This chapter deals with the commonly used welding processes namely manual metal arc, submerged arc, gas shielded tungsten arc, gas shielded metal arc and oxyacetylene process only. For other processes, the requirements are to be mutually agreed between the manufacturer/contractor and the authorized inspector.

### 7.1 Manufacturer's Responsibilities

**7.1.1** Each manufacturer or contractor is responsible for the welding done by his organization. He shall establish the procedure and conduct the tests required in this section to qualify the welding procedures and the performance of welders and welding operators who apply these procedures.

**7.1.2** When a manufacturer or contractor establishes proof satisfactory to the inspector that he has previously made successful procedure qualification tests in accordance with a recognized standard, such a firm shall be exempted from the necessity of requalification provided all the requirements of this section are met.

**7.1.3** The parameters applicable to the welding he performs shall be listed in a document known as 'Welding Procedure Specification' (WPS) ( see Appendix H of the code for suggested WPS format ). The WPS shall be qualified by welding test coupons and testing the specimens as required in this section. The WPS shall describe all of the essential and non-essential variables of 7.3 and 7.4 for each welding process used in WPS. The WPS may be used to provide direction to the welder or welding operator and may include any other information that may be helpful in making a weldment. The welding data and test results

shall be recorded in a document known as 'Procedure Qualification Record' (PQR) (see Appendix H for suggested PQR format). The PQR is a record of what happened during a particular welding test and changes to PQR are not permitted unless requalified by the manufacturer or contractor.

**7.1.4** Test plate is to be welded by the manufacturer or contractor and he may subcontract testing.

## **7.2 Test Positions**

**7.2.1** All test welds for welding procedure qualification shall be carried out as groove welds in any of the positions described in the following paragraphs, except that the angular deviation shall be within  $\pm 15$  degree from the specified horizontal and vertical planes and  $\pm 5$  degree from the specified inclined plane.

**7.2.2** The following are the basic positions for welding (see Fig. 7.1).

**7.2.2.1 Flat position** — Plate in horizontal plane with the weld metal deposited from above. Pipe with its axis horizontal and rolled during welding so that the weld metal is deposited from above (see Fig. 7.1 A-1 and Fig. 7.1 A-2).

**7.2.2.2 Horizontal position** — Plate in vertical plane with the axis of weld horizontal:

Pipe with its axis vertical and the axis of weld in horizontal plane:

Pipe shall not be rotated during welding (see Fig. 7.1 B-1 and Fig. 7.1 B-2).

**7.2.2.3 Vertical position** — Plate in vertical plane with the axis of weld vertical (see Fig. 7.1C).

**7.2.2.4 Overhead position** — Plate in horizontal position with the weld metal deposited from underneath (see Fig. 7.1D).

**7.2.2.5 Multiple position horizontal** — Pipe with its axis horizontal and the welding groove in a vertical plane. Welding shall be done without rotating the pipe so that weld metal is deposited in multiple position (see Fig. 7.1E).

**7.2.2.6 Multiple position inclined** — Pipe with its axis inclined at  $45^\circ$  to horizontal. Welding shall be done without rotating the pipe (see Fig. 7.1F).

**7.3 Essential Variables** — Procedure qualification is required whenever there is a change in the welding condition which will affect the properties of the weldment, namely, change in the material group, welding process, filler metal, preheat or postweld heat treatment. Retesting is also required beyond the range qualified and for any change in position, for low temperature vessels.

**7.3.1 Base Metal** — Procedure qualification required in the following cases.

**7.3.1.1** A change from a base metal listed under one metal group given in Table 6.3 of the code to the metal listed under another with the same metal group, the procedure qualified on metal with certain minimum tensile strength does not qualify for welding metal with higher specified tensile strength.

**7.3.1.2** When a joint is made between two base metals that have different material groups even though qualification testing have been made for each of the two base metals welded to itself.

**7.3.1.3** When the thickness of base metal is beyond the range qualified in accordance with Table 7.1 and 7.3.1.5 and 7.3.6.3.

**7.3.1.4** Any major change in weld preparation details for low temperature operation and single-sided joint.

**7.3.1.5** For the short circuiting transfer made of the gas shielded arc process when the thickness exceeds 1.1 times the thickness of test coupon.

**7.3.2 Welding Process** — For each welding process a new WPS is required and each process shall be qualified. This may be qualified either separately or in combination with other processes. For multi-processes, the qualified thickness of each process shall not be additive in determining the thickness qualified.

**7.3.3 Filler Metal** — The filler metal classification to be used on the production weld should be same as the one used for procedure qualification. Any appreciable variation in the filler metal classification from the one used for the procedure will require requalification. In case of single welded joint, a change of more than one-third in the diameter of electrode for root run of manual metal arc welding requires requalification for low temperature operation.

**7.3.4 Position (Low Temperature Vessels only)** — The change of position is an essential variable for low temperature vessels only and requires requalification. However, if the position qualified is vertical up, it qualifies for other positions also. In vertical position, a change from stringer bead to weave bead calls for requalification.

**7.3.5 Preheat** — The minimum preheat temperature shall be specified in WPS. A decrease of more than 50 degree celsius and an increase of more than 100 degree celsius for low temperature and stainless steel jobs from the minimum specified shall require requalification.

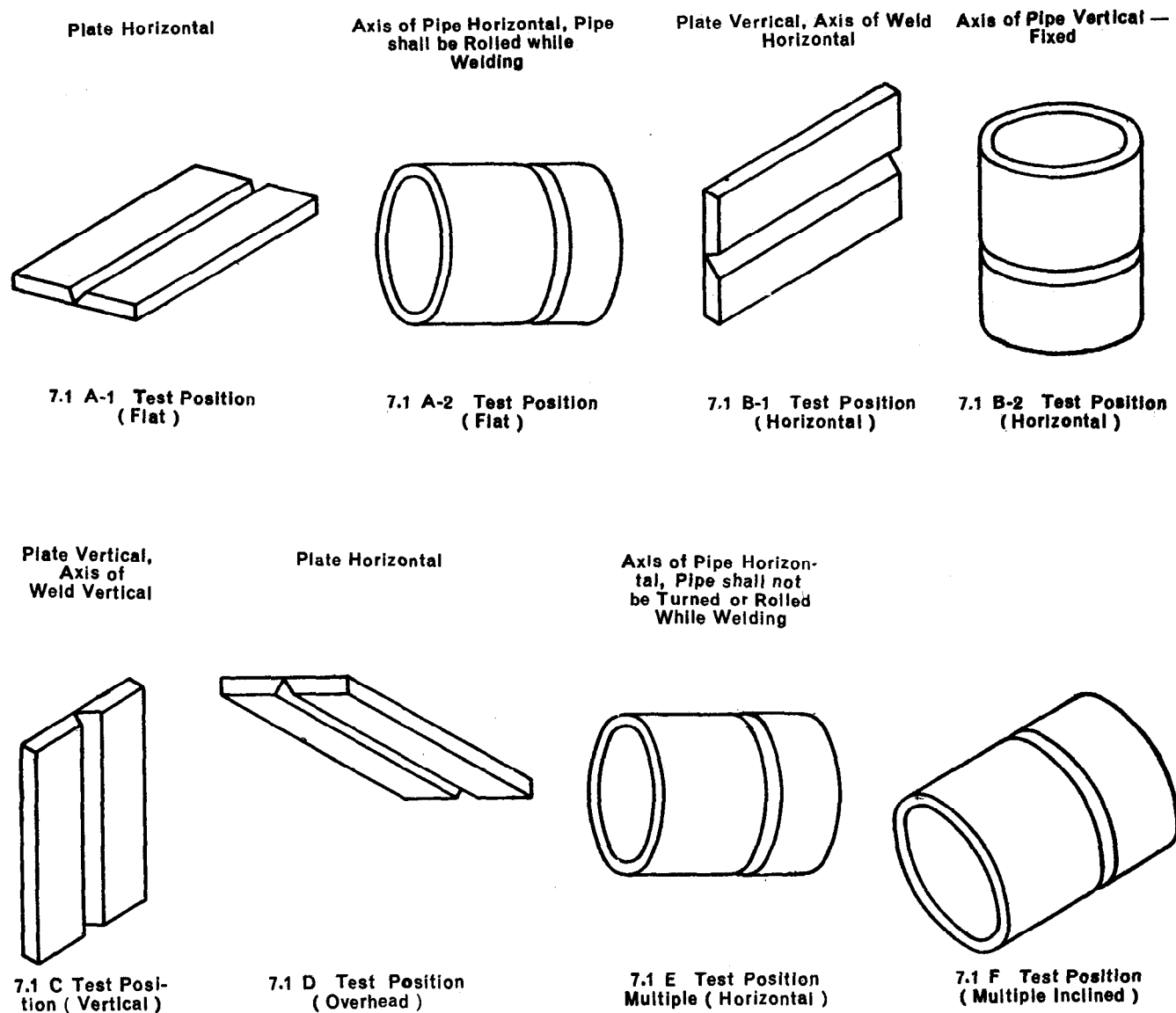


FIG. 7.1 POSITIONS OF TEST PLATES OR PIPES FOR WELDER QUALIFICATION AND PERFORMANCE OF BUTT WELDS

**TABLE 7.1 GROOVE-WELD PROCEDURE QUALIFICATION — THICKNESS LIMITS AND TEST SPECIMENS**

( Clauses 7.3.1.3, 7.5.1, 7.6.9, 7.7.2 and 7.7.4 )

| THICKNESS 'T'<br>OF TEST<br>COUPON<br>(see NOTE 1.1)<br>mm | RANGE OF<br>THICKNESS OF<br>BASE METAL<br>QUALIFIED<br>( see NOTES 1<br>AND 7 ) |           | RANGE OF<br>THICKNESS<br>'t' OF<br>DEPOSITED<br>WELD<br>METAL<br>QUALIFIED<br>Max (see<br>Notes 1<br>AND 7 )<br>mm | TYPES AND NUMBER OF TESTS REQUIRED                |                              |                                                                            |                                 |              |                                                                      |                                                                            |                                                                             |
|------------------------------------------------------------|---------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------|----------------------------------------------------------------------------|---------------------------------|--------------|----------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------|
|                                                            | Min<br>(see Note 5)<br>mm                                                       | Max<br>mm |                                                                                                                    | Reduced<br>Section<br>Tension<br>( see<br>7.5.2 ) | All Weld<br>Metal<br>Tension | Trans-<br>verse<br>Face<br>Bend<br>( see 7.5.4<br>and Notes 1.2<br>and 3 ) | Trans-<br>verse<br>Root<br>Bend | Side<br>Bend | Notched Bar<br>Weld Metal<br>( see 7.5.1.4<br>and<br>Notes 4 and 6 ) | Impact<br>Heat<br>Affected<br>Zone<br>( see 7.5.1.4<br>and Notes 4 and 6 ) | Macro<br>and<br>Hardness<br>( see 7.5.1.5<br>and 7.5.1.6<br>and<br>Note 6 ) |
|                                                            |                                                                                 |           |                                                                                                                    |                                                   |                              |                                                                            |                                 |              |                                                                      |                                                                            |                                                                             |
|                                                            |                                                                                 |           |                                                                                                                    |                                                   |                              |                                                                            |                                 |              |                                                                      |                                                                            |                                                                             |
| (1)                                                        | (2)                                                                             | (3)       | (4)                                                                                                                | (5)                                               | (6)                          | (7)                                                                        | (8)                             | (9)          | (10)                                                                 | (11)                                                                       | (12)                                                                        |
| Below 3                                                    | T                                                                               | 3         | 3                                                                                                                  | 1                                                 | —                            | 2                                                                          | 2                               | —            | —                                                                    | —                                                                          | 1                                                                           |
| 3 up to and<br>including 10                                | 3                                                                               | 2T        | 2t                                                                                                                 | 1                                                 | —                            | 2                                                                          | 2                               | —            | 3                                                                    | 3                                                                          | 1                                                                           |
| Over 10 but<br>less than 20                                | 5<br>(see Note 7)                                                               | 2T        | 2t                                                                                                                 | 1                                                 | 1                            | 2                                                                          | 2                               | (see Note 2) | 3                                                                    | 3                                                                          | 1                                                                           |
| 20 and less<br>than 40                                     | 5                                                                               | 2T        | 2t                                                                                                                 | 1                                                 | 1                            | —                                                                          | —                               | 4            | 3                                                                    | 3                                                                          | 1                                                                           |
| 40 and above                                               | 5                                                                               | 200       | 200 when<br>t ≥ 40                                                                                                 | 1                                                 | 1                            | —                                                                          | —                               | 4            | 3                                                                    | 3                                                                          | 1                                                                           |

NOTE 1 — When the groove is filled using a combination of welding processes and/or welding procedures with the same process ( one welding process with a different combination of essential variable ).

NOTE 1.1 — The thickness *t* of the deposited weld-metal for each welding procedure shall be determined and used in the thickness *t* of deposited weld metal column. The test coupon thickness *T* is applicable for the base metal for each welding procedure and shall be used in the range of thickness of base metal column.

NOTE 1.2 — The deposited weld metal of each welding process and of each welding procedure shall be included on the tension side of the bend when these test samples are used.

NOTE 1.3 — Each welding process and each welding procedure qualified in this combination manner may be used separately, only within the same essential variables and within the thickness limits described in this table.

NOTE 2 — Side-bend test may be substituted for the required face and root bend tests, when thickness *T* is over 10 mm but less than 20 mm.

NOTE 3 — Longitudinal face and root bend tests may be substituted for the required transverse face, root and side bend tests, when the bending properties of the two base metals or the weld metal and the base metal differ markedly ( see 7.5.4 and 7.5.5 ).

NOTE 4 — Out of three specimens for notched bar impact test for weld metal as well as heat-affected zone ( HAZ ), two specimens shall contain the face side of the joint and one specimen the root side of the joint. These form one set of specimens ( see 7.5.6 ).

NOTE 5 — See also 7.3.1.5 and 7.3.6.3 for further limitations on range of thickness of base metal qualified. For oxy-acetylene welding the test piece thickness shall be the maximum thickness for which the procedure qualification is valid.

NOTE 6 — Notch bar impact and hardness tests and macro examination are applicable to class I vessels only ( see Table 1.1 ).

NOTE 7 — For low temperature application, the minimum base metal thickness qualifies ' *T* ' for thickness below 16 mm and 16 mm for thickness 16 mm and above.

### 7.3.6 Post Weld Heat Treatment

**7.3.6.1 Post weld heat treatment (PWHT)** is a variable and requires qualification under each of the conditions mentioned below. The test coupon shall be subjected to heat treatment essentially equivalent to that encountered in the base metal and fabrication of weldments, including at least 80 percent of the aggregate times at that temperature. This 80 percent time limitation is not applicable to post weld heat treatment when it is conducted below the lower transformation temperature:

- a) No PWHT;
- b) PWHT below lower transformation temperature (for example, tempering);
- c) PWHT above upper transformation temperature (for example, normalizing);
- d) PWHT shows upper transformation temperature followed by heat treatment below the lower transformation temperature (for example, normalizing or quenching followed by tempering); and
- e) PWHT between the upper and lower transformation temperatures.

**7.3.6.2** In case the weld is for low temperature application a change from post weld heat treatment and time range in accordance with 7.3.6.1 requires requalification.

**7.3.6.3** For the test coupons receiving a post weld heat treatment in which either of the critical temperatures is exceeded the maximum qualified thickness for production welds is 1.1 times the thickness of test coupon.

**7.3.7 Shielding Gas** — In case of gas shielded process a change from a single shielding gas to another shielding gas or to a mixture of shielding gases or a change in the composition of the shielding gas mixture or omission of shielding gas will require requalification. In case of non-ferrous metals, a change of more than 15 percent in the gas flow rate requires requalification.

**7.3.8 Electrical Characteristics** — In case of gas shielded arc process for non-ferrous materials, a change from dc to ac or *vice versa* and in dc welding, change in electrode polarity requires requalification. A change from spray arc, globular arc or pulsating arc to short circuiting arc or *vice versa* requires requalification.

**7.4 Non-essential Variables** — Variables other than above arc non-essential variables including change of location and does not require fresh procedure qualification for metal arc welding, submerged arc welding and gas shielded arc welding. The WPS may alone be revised to show the non-essential variable where necessary. For other welding processes the essential variables will be drawn in consultation with the Inspection Authority.

## 7.5 Tests

**7.5.1** The following tests are required for procedure qualification (see Table 7.1).

**7.5.1.1 Reduced section tensile test** — One test is done to determine the ultimate tensile strength of the groove welded joint.

**7.5.1.2 All weld metal tensile test** — One test is done to determine the ultimate tensile strength and elongation of weld metal.

**7.5.1.3 Guide bend tests** — Guide bend tests as described below are done to determine soundness and ductility of groove weld joints:

- a) One transverse root bend up to 10 mm thickness (see 7.5.4);
- b) One transverse face bend up to 10 mm thickness (see 7.5.4);
- c) Two transverse side bends above 10 mm thickness (see 7.5.4);
- d) One longitudinal root bend instead of transverse root bend (see 7.5.5); and
- e) One longitudinal face bend instead of transverse face bend (see 7.5.5).

**7.5.1.4 Notched bar impact test** — One set of three charpy V-notch impact test are done in accordance with 7.5.5 to determine notch toughness property of the groove welds (see also 7.5.6).

**7.5.1.5 Macro test** — One test is done in accordance with 7.5.6 to check the complete fusion of the groove weld.

**7.5.1.6 Hardness test** — This is done in accordance with 7.5.7 to check the hardness of weld and heat affected zone (HAZ).

**7.5.2 Reduced Section Tensile Test** — One reduced section tensile test on a test specimen cut transverse to the weld is to be done [see IS : 1608-1972 Method of tensile testing of steel products (first revision)] as follows.

**7.5.2.1** A single specimen of full thickness shall be used for thickness up to and including 25 mm for plates (see Fig. 7.2A) and pipes (see Fig. 7.2B);

**7.5.2.2** For thicknesses above 25 mm single or multiple specimen may be used as necessary for plates and pipes.

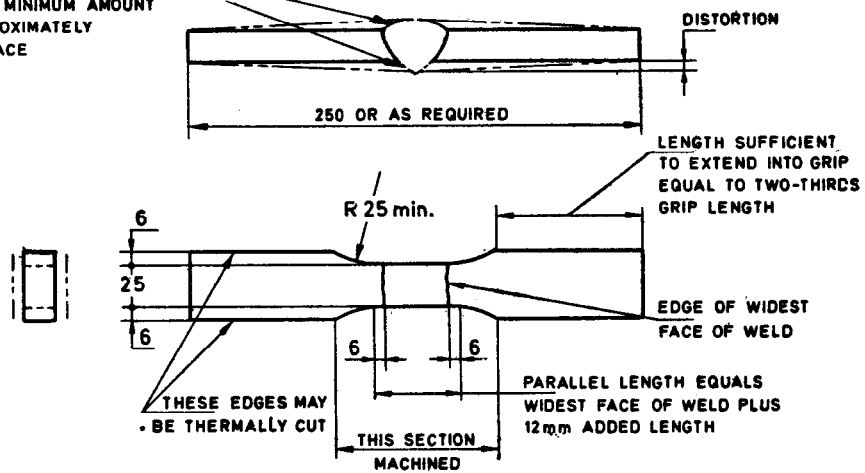
**7.5.2.3** When multiple specimens are used the specimen shall be cut so as to represent the full thickness of the weld at the location. All the specimens shall represent the full thickness of the weld at one location and each specimen shall be tested.

**7.5.2.4** For small pipes (such as less than 73 mm O.D.) reduced section specimen as per Fig. 7.2C may be used.

**7.5.2.5** Alternatively full section specimen as per Fig. 7.3 may be used.

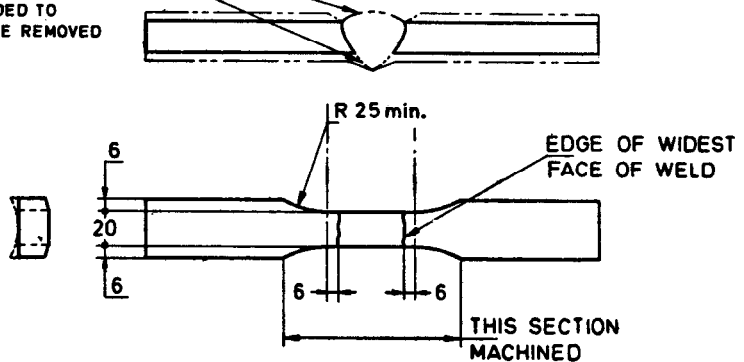
**7.5.3 All Weld Metal Tensile Test** — One all weld metal tensile test on a test specimen taken along the weld is to be done (see IS : 1608-1972) on a specimen prepared as shown in Fig. 7.4 (see 8.5.6) as follows:

WELD REINFORCEMENT SHALL BE MACHINED FLUSH WITH BASE METAL. MACHINE MINIMUM AMOUNT TO OBTAIN APPROXIMATELY PARALLEL SURFACE

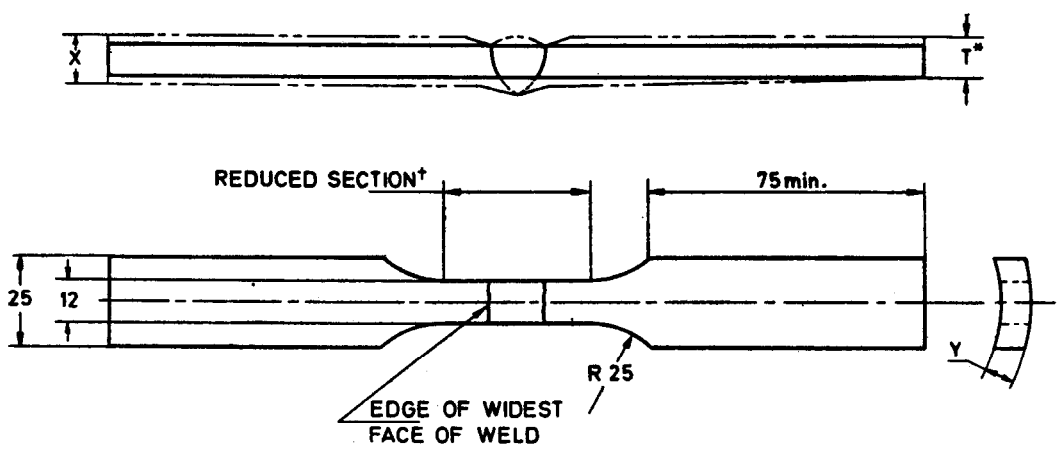


7.2A Reduced Section-Plate

GRIND OR MACHINE MINIMUM AMOUNT NEEDED TO OBTAIN PLANE PARALLEL FACE OVER 20mm WIDE REDUCED SECTION. NO MORE MATERIAL THAN IS NEEDED TO PERFORM TEST SHALL BE REMOVED



7.2B Reduced Section-Pipe



\*The weld reinforcement shall be ground or machined so that the weld thickness does not exceed the base metal thickness 'T'. Machine minimum amount to obtain approximately parallel surfaces.  
†The reduced section shall not be less than the width of the weld plus 2T.

7.2C Reduced Section-Pipe < 75 mm OD

All dimensions in millimetres.

FIG. 7.2 REDUCED SECTION TENSILE TEST SPECIMEN

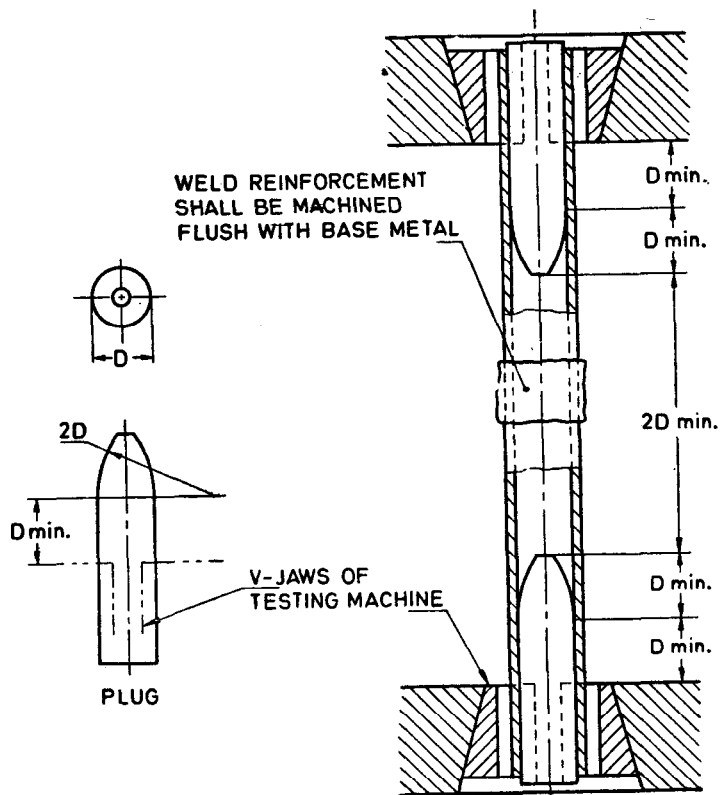
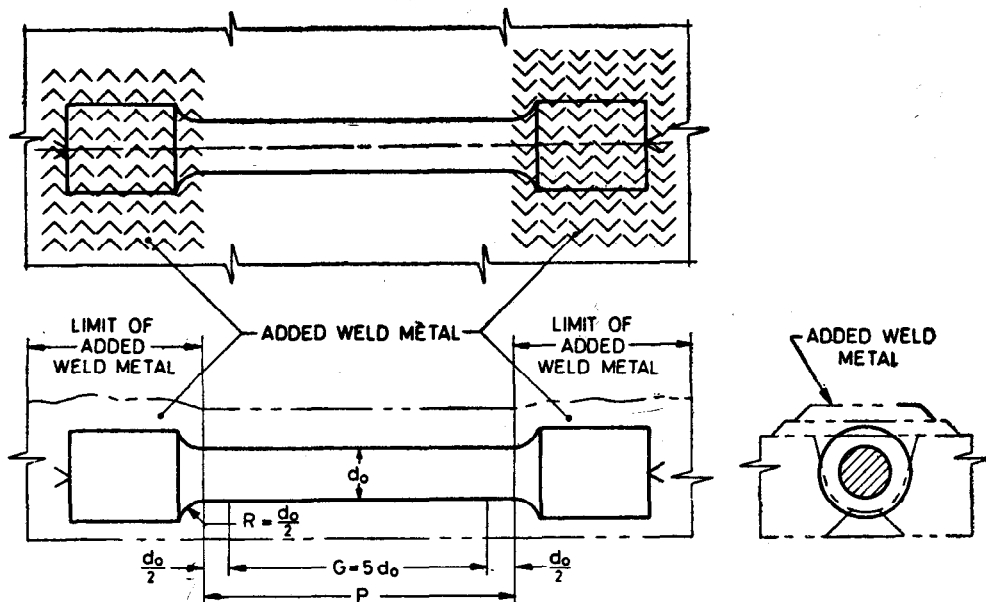


FIG. 7.3 FULL SECTION TENSILE TEST SPECIMEN — SMALL PIPE



NOTE —  $d_o$  Maximum possible diameter, need not be more than 16 mm.

FIG. 7.4 ALL WELD METAL TENSILE TEST SPECIMEN

**7.5.3.1** The diameter of the test specimen is the maximum possible diameter and this need not be more than 16 mm.

**7.5.3.2** The test should include tensile strength, yield strength and elongation.

**7.5.3.3** For design temperature above 100 degree celsius the test shall include elevated temperature testing.

**7.5.3.4** In case of multiple process or multiple procedure the specimen are to be prepared to represent each process/procedure. The location and size of specimen are selected in consultation with the Inspection Authority.

**7.5.3.5** For pipe instead of one all weld metal tensile test, one extra transverse tensile test is to be done ( see Fig. 7.9 ).

**7.5.4 Transverse Bend Test** — Two transverse root bend tests and two face bend tests are to be done on test specimens cut transverse to the weld ( see 8.5.9 ). The diameter of former and distance between supports shall be as given in Table 8.2.

**7.5.4.1** The specimens are bent such a way that the root surface becomes convex surface in one set and the face surface becomes convex surface in another set ( see Fig. 7.5A and Fig. 7.5B ).

**7.5.4.2** When the thickness of plate exceeds 10 mm, face bend and root bend tests are to be substituted by two side bend tests ( see Fig. 7.5C ).

**7.5.5 Longitudinal Bend Test** — Transverse bend tests may be substituted with one face bend and one root longitudinal bend test ( see Fig. 7.6 ), when the weld metal and the parent metal differ markedly in bending properties either between dissimilar parent metals or weld metal and parent metal. The tests shall be done as per transverse bend test procedure ( see 7.5.4 ).

**7.5.6 Notched Bar Impact Test** — One test of three V-notch impact test specimens transverse to the weld is to be tested as per 8.5.8 of the code. In case of welds for low temperature application, three more impact test specimens shall be taken from the heat affected zone and tested.

**7.5.7 Macro Test** — One macro test on a specimen cut transverse to the weld is to be done to determine the complete fusion and free from cracks, as per the code ( see 8.5.11 ).

**7.5.8 Hardness Test** — Hardness test is to be done on the macro test specimen along the cross section of the weld and the heat affected zone. The hardness shall be assessed by a HV indenter of not more than 30 kgf load,

**7.5.9 Preparation of Test Coupon** — The base metal and filler metal shall be as per 7.7.

**7.5.10 Order of Removal of Test Specimens** — The test specimens are removed from the test coupon as detailed below:

a) Where transverse bend tests are required,

the test specimens are removed as shown in Fig. 7.7.

b) Where longitudinal bend tests are required, the test specimens are removed as shown in Fig. 7.8.

c) Where test is done on a pipe, the test specimens are removed as shown in Fig. 7.9.

**7.6 Acceptance Criteria** — In order to pass the test, specimens shall have the following minimum requirements.

#### **7.6.1 Transverse Tensile Test**

**7.6.1.1** Specified minimum tensile and yield strength of base metal.

**7.6.1.2** In case of two different metals, specified minimum strength of the weaker of the two.

**7.6.1.3** If the specimen breaks outside the fusion line, the test shall be accepted as meeting the requirement provided the strength is more than 95 percent of the minimum tensile strength of the base metal.

#### **7.6.2 All Weld Metal Test**

**7.6.2.1** The tensile strength and/or yield strength shall not be less than the corresponding specified minimum values for the parent metal.

**7.6.2.2** The elongation shall not be less than 80 percent of the minimum elongation specified for the base metal corrected for the gauge length if different from 5  $d_0$ .

**7.6.2.3** For elevated temperature test, the values shall not be less than corresponding values for the base metal at the appropriate temperature.

#### **7.6.3 Bend Tests**

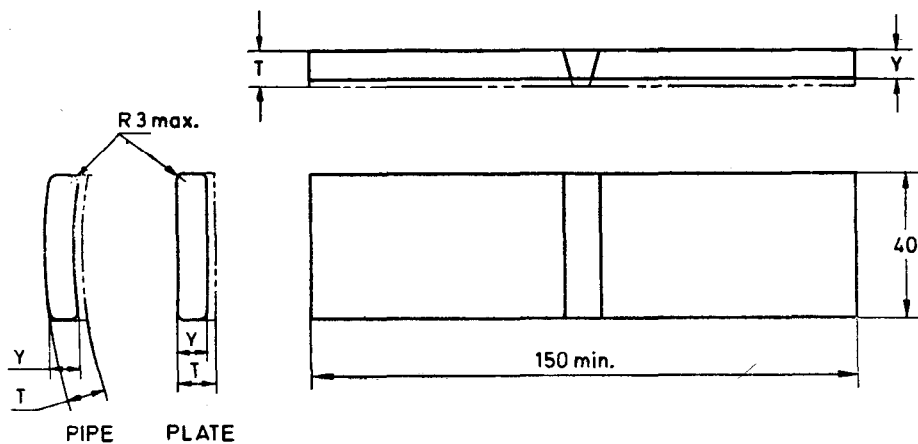
**7.6.3.1** The weld and heat affected zone of a transverse bend specimen shall be completely within the bent portion of the specimen after testing.

**7.6.3.2** The specimen shall have no open defect in the weld or heat affected zone exceeding 3 mm measured in any direction on the convex surface of the specimen.

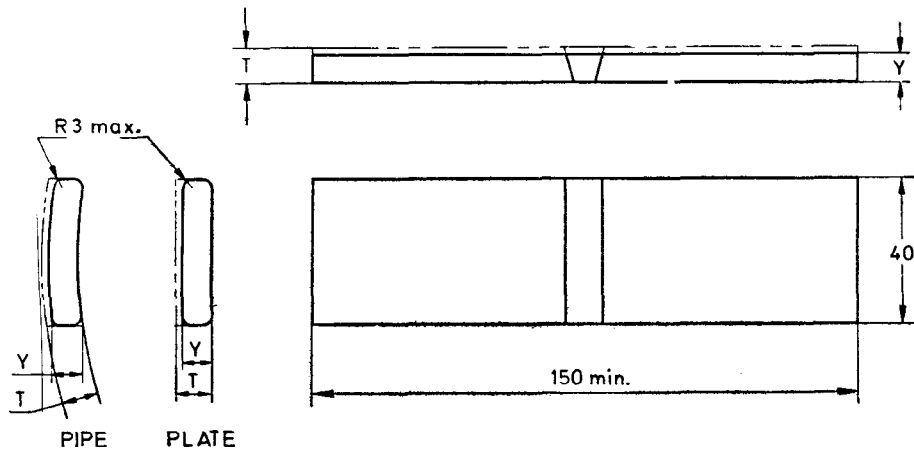
**7.6.3.3** Cracks occurring on the corners of the specimen during the testing shall not be considered unless there is definite evidence that they result from slag inclusions or other internal defects.

**7.6.4 Impact Test** — The minimum average results from the impact test shall be 2.6 kgf/cm<sup>2</sup> and the minimum value for any test specimen shall be 2.0 kgf/cm<sup>2</sup>.

**7.6.5 Macro Examination** — Visual examination of the cross section of the weld metal and heat affected zone ( HAZ ) shall show complete fusion and shall be free from cracks.



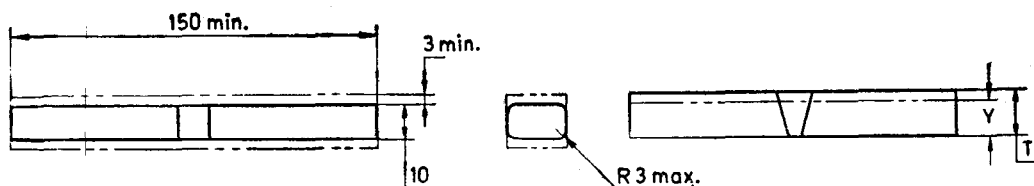
7.5A Face-Bend Specimen — Plate and Pipe



7.5B Root Bend Specimen — Plate and Pipe

*Special Note for 7.5A and 7.5B*

1.  $Y = T$  when  $T$  is 1.5 mm to 10 mm.

*Special Notes for 7.5C*

1. When specimen thickness  $T$  exceeds 20 mm, use one of the following:
  - 1.1 Cut specimen into multiple test coupons  $T$  of approximately equal dimensions (10 mm to 40 mm).
  - $T$  = tested specimen thickness when multiple specimens are taken from one coupon.
  - 1.2 The specimen may be bent at full width.
2.  $T = T$  when  $T$  is 10 to 20 mm.

7.5C Side Bend Specimen — Plate and Pipe

NOTE 1 — Weld reinforcement and backing strip or backing ring, if any shall be removed flush with the surface of the specimen.

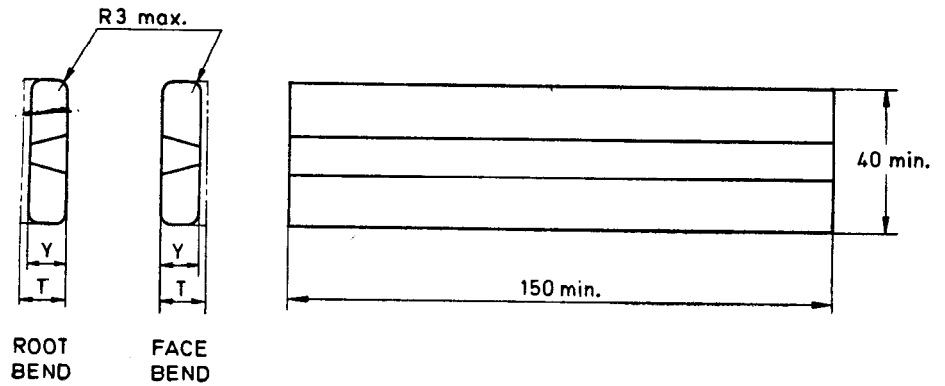
NOTE 2 — If the pipe being tested is 100 mm nominal diameter or less, the width of the bend specimen may be as follows:

- 2.1 20 mm for diameters 50 mm to 100 mm including
- 2.2 10 mm for diameters less than 50 mm to 10 mm including
- 2.3 Alternatively, if the pipe is 25 mm nominal dia or less, the width may be that obtained by cutting the pipe into quarter sections, less an allowance for cutting. These specimens are not required to have one surface machined flat as shown.

NOTE 3 — If the surfaces of bend specimen are gas cut, removal by machining or grinding of not less than 3 mm from the surface shall be required.

All dimensions in millimetres.

FIG. 7.5 TRANSVERSE BEND TEST SPECIMENS



$Y = T$  WHEN  $T$  IS 1.5mm TO 10mm

$Y = 10\text{mm}$  WHEN  $T > 10\text{mm}$

FIG. 7.6 LONGITUDINAL BEND TEST SPECIMEN

|                       |                                                                                     |                  |
|-----------------------|-------------------------------------------------------------------------------------|------------------|
| DISCARD               |                                                                                     | THIS PIECE       |
| REDUCE SECTION        |                                                                                     | TENSILE SPECIMEN |
| ROOT BEND (SIDE BEND) |                                                                                     | SPECIMEN         |
| FACE BEND (SIDE BEND) |                                                                                     | SPECIMEN         |
| ALL-WELD METAL        |  | TENSILE SPECIMEN |
| NOTCHED BAR           |                                                                                     | IMPACT SPECIMEN  |
| NOTCHED BAR           |                                                                                     | IMPACT SPECIMEN  |
| NOTCHED BAR           |                                                                                     | IMPACT SPECIMEN  |
| MACRO TEST            |                                                                                     | SPECIMEN         |
| DISCARD               |                                                                                     | THIS PIECE       |

FIG. 7.7 ORDER OF REMOVAL OF TEST SPECIMENS FROM PLATE-TRANSVERSE BEND PROCEDURE QUALIFICATION

|                 |                                                                                    |                    |
|-----------------|------------------------------------------------------------------------------------|--------------------|
| DISCARD         |                                                                                    | THIS PIECE         |
| LONGITUDINAL    |                                                                                    | FACE BEND SPECIMEN |
| REDUCED SECTION |                                                                                    | TENSILE SPECIMEN   |
| LONGITUDINAL    |                                                                                    | ROOT BEND SPECIMEN |
| MACRO TEST      |                                                                                    | SPECIMEN           |
| ALL WELD METAL  |  | TENSILE SPECIMEN   |
| NOTCHED BAR     |                                                                                    | IMPACT SPECIMEN    |
| NOTCHED BAR     |                                                                                    | IMPACT SPECIMEN    |
| NOTCHED BAR     |                                                                                    | IMPACT SPECIMEN    |
| DISCARD         |                                                                                    | THIS PIECE         |

FIG. 7.8 ORDER OF REMOVAL OF TEST SPECIMENS,  
PLATE-LONGITUDINAL BEND PROCEDURES  
QUALIFICATION

**7.6.6 Hardness Test** — The maximum Vicker's hardness value of the weld metal and heat affected zone (HAZ) shall not exceed the following values (see also Table 6.3).

**7.6.6.1** 225 HV for steels of metal group 0, 1a and 1b.

**7.6.6.2** 240 HV for steels of metal group 2.

**7.6.6.3** 250 HV for steels of metal group 3 and 4.

**7.6.7 Acceptance of Procedure Qualification Record (PQR)** — The results of the test and the examination of the test coupons shall satisfy the above requirements for acceptance.

**7.6.8 Records** — Records of all tests (see 7.1.3) shall be kept by the manufacturer for a period of at least 5 years after the inspection of the pressure vessel and shall be available for the Inspecting Authority for examination when required. Proforma for keeping records is given in Appendix H.

#### 7.6.9 Qualification of Welding Procedure Specification (WPS)

**7.6.9.1** Qualification of plate qualifies for the pipe and *vice versa*.

**7.6.9.2** Qualification on a groove weld qualifies the fillet weld of all sizes on all base metal thicknesses as permitted in Table 7.1.

**7.6.9.3** Qualification on a groove weld qualifies partial penetration groove welds for the thickness of deposited weld metal qualified as indicated in Table 7.1.

**7.6.9.4** The range of base metal thickness and weld metal thickness qualified as is indicated in Table 7.1.

**7.6.9.5** WPS qualified on groove welds shall be applicable for production welds between dissimilar base metal thicknesses provided:

- The thickness of thinner member shall be within range permitted.
- The production joints shall be within the thickness range permitted in Table 7.1.
- Alternatively, the maximum thickness permitted in Table 7.1 may be adopted for thicker member provided the qualification was made on the base metal of thickness 40 mm or more.
- More than one PQR may be required to qualify for some dissimilar thickness combination.

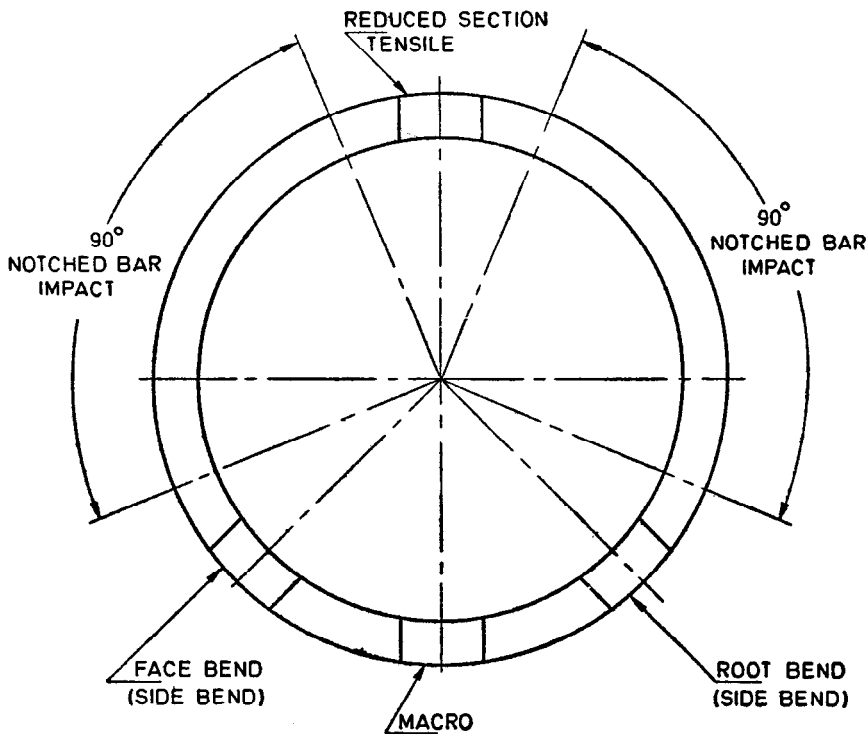
**7.7 Preparation of Test Coupon** — The base metal and filler metal shall be one or more of those listed in the WPS. The dimension of the test coupon shall be sufficient to provide the required test specimens. The base metal may consist of either a plate or a pipe. Qualifications of plate also qualifies the pipe and *vice versa*.

**7.7.1** The order of removal of test specimens for plates and pipes are shown in Fig. 7.7 to 7.9.

**7.7.2** The test coupon shall qualify the thickness ranges of both base metal and the deposited weld metal to be used in production. Limits of qualification shall be in accordance with Table 7.1.

**7.7.3** The WPS made as above on groove weld qualifies the fillet weld for all fillet sizes on all base metal thicknesses.

**7.7.4** The WPS made as above on groove weld qualifies partial penetration groove welds for the deposited metal thickness indicated in Table 7.1.



NOTE — Weld extra test piece to accommodate one extra transverse tensile specimen in place of all weld metal tensile and to accommodate three impact specimens, if required.

FIG. 7.9 ORDER OF REMOVAL OF TEST SPECIMENS, PIPE-PROCEDURE QUALIFICATION

## 7.8 Welder's Performance Qualification

**7.8.1** Each manufacturer or contractor shall be responsible for conducting tests to qualify the performance of the welders or welding operators in accordance with one of this welding procedure specification (WPS). The welders or welding operators shall be tested under the supervision and control of the manufacturer or contractor. The welding performed by the welder or welding operator in another organization is not to be considered.

**7.8.2 Qualification Tests** — The performance qualification tests are intended to determine the ability of the welder or welding operator to make sound welds. The manufacturer or contractor shall qualify each welder or welding operator for each welding process to be used in production welding. The performance tests can be either for groove weld or for fillet weld.

**7.8.2.1** The welder or welding operator who welds the PQR test coupons is also qualified within the limits of the performance qualification listed in this section (see 7.8.5).

**7.8.2.2** A welder or welding operator qualified to weld in accordance with one qualified WPS is also qualified to weld in accordance with other qualified WPS using the same welding process within the limits of variables listed in this section (see 7.8.7).

**7.8.2.3** Each qualified welder or welding operator shall be assigned an identification which shall be used to identify the work of that welder or welding operator.

**7.8.2.4** The record of welder or welding operator performance tests shall be kept by the manufacturer. This shall include the welding variable, the type of test, tests results and ranges qualified for each welder and welding operator. Pro-forma for keeping records is given in Appendix H.

## 7.8.3 Groove Welds

**7.8.3.1 Mechanical test** — The type and number of test specimens required shall be in accordance with Table 7.2. The order of removal of test specimen is shown in Fig. 7.10. The acceptance shall be in accordance with 7.6.

**7.8.3.1.1 Mechanical testing as given in Table 7.2** may be substituted by radiography on a test bed of 225 mm length. Acceptance standard for radiography shall be as per Class I vessels. Qualification by radiography is not permitted for gas metal arc welding.

**7.8.3.2 Test coupon for pipes** — The test coupons for pipes shall be removed as shown in Fig. 7.11 for mechanical testing. The acceptance shall be in accordance with 7.6.

**TABLE 7.2 GROOVED-WELD PERFORMANCE QUALIFICATIONS THICKNESS LIMITS AND TEST SPECIMENS**

( Clause 7.8.3.1 )

| THICKNESS $T$ OF<br>THE TEST<br>COUPON WELDED<br>( see NOTE 1 )<br>mm | THICKNESS $t$ OF<br>DEPOSITED WELD<br>METAL QUALIFIED<br>( see 7.8.6 AND<br>NOTE 2 )<br>mm | TYPE AND NUMBER OF TEST REQUIRED |                                    |           |       |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------|------------------------------------|-----------|-------|
|                                                                       |                                                                                            | Side Bend                        | Face Bend<br>( see Notes 3 and 4 ) | Root Bend | Macro |
| Up to 10                                                              | $2t$                                                                                       | ( see Note 5 )                   | 1                                  | 1         | 1     |
| 10 but less<br>than 20                                                | $2t$                                                                                       | 2                                | —                                  | —         | 1     |
| 20 and above                                                          | Maximum to be<br>welded                                                                    | 2                                | —                                  | —         | 1     |

NOTE 1 — The entire thickness of groove joint test coupon shall be filled with deposited weld metal.

NOTE 2 — Two or more pipe test coupons of different thicknesses may be used to determine the deposited weld metal thickness qualified and that thickness may be applied to production welds to the smallest diameter for which the welder is qualified.

NOTE 3 — A total of four specimens is required to qualify for multiple positions as prescribed in Fig. 7.11B.

NOTE 4 — Face and root-bend tests may be used to qualify combination test of:

4.1 One welder using two welding processes; or

4.2 Two welders using the same or a different welding process.

NOTE 5 — Two side bend tests may be substituted for the required face and root-bend tests.

|                          |  |            |
|--------------------------|--|------------|
| DISCARD                  |  | THIS PIECE |
| ROOT BEND<br>(SIDE BEND) |  | SPECIMEN   |
| FACE BEND<br>(SIDE BEND) |  | SPECIMEN   |
| MACRO TEST               |  | SPECIMEN   |
| DISCARD                  |  | THIS PIECE |

7.10A Transverse Bend

|                           |  |            |
|---------------------------|--|------------|
| DISCARD                   |  | THIS PIECE |
| LONGITUDINAL<br>FACE BEND |  | SPECIMEN   |
| LONGITUDINAL<br>ROOT BEND |  | SPECIMEN   |
| MACRO<br>TEST             |  | SPECIMEN   |
| DISCARD                   |  | THIS PIECE |

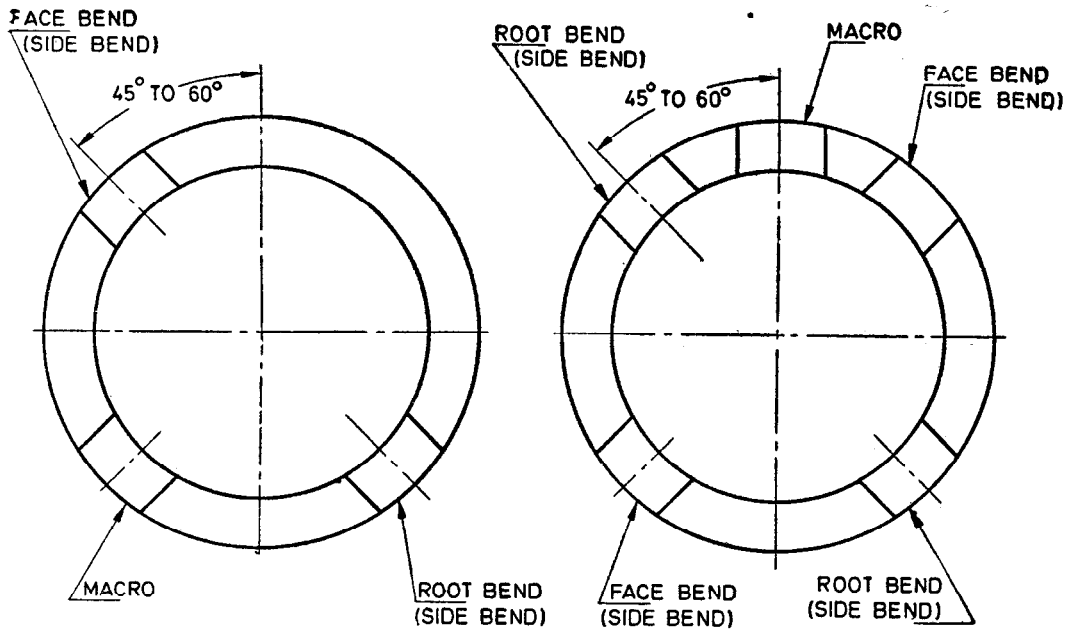
7.10B Longitudinal Bend

**FIG. 7.10 ORDER OF REMOVAL OF TEST SPECIMEN, PLATE — PERFORMANCE QUALIFICATION****7.8.4 Fillet Welds**

**7.8.4.1 Mechanical test** — For fillet welds, fracture test and macro examination shall be done. The dimension and preparation of test specimen shall be in accordance with Fig. 7.12 for plate and Fig. 7.13 for pipe. The test specimen shall not contain any visible cracks.

**7.8.4.2** The specimen for fracture test is

taken as shown in Fig. 7.12 and Fig. 7.13 and shall be loaded laterally in such a way that the root of the weld is in tension. The load shall be steadily increased until the specimen fractures or bends flat upon itself. The fractured surface shall show no evidence of cracks or incomplete root fusion and the sum of the length of inclusion and porosity visible on the fractured surface shall not exceed 10 mm for plate and 10 percent of the quarter section for pipe.



7.11A Flat or Horizontal  
7.11B Multiple Horizontal and Multiple-Inclined  
FIG. 7.11 ORDER OF REMOVAL OF TEST SPECIMENS, PIPE-PERFORMANCE QUALIFICATION

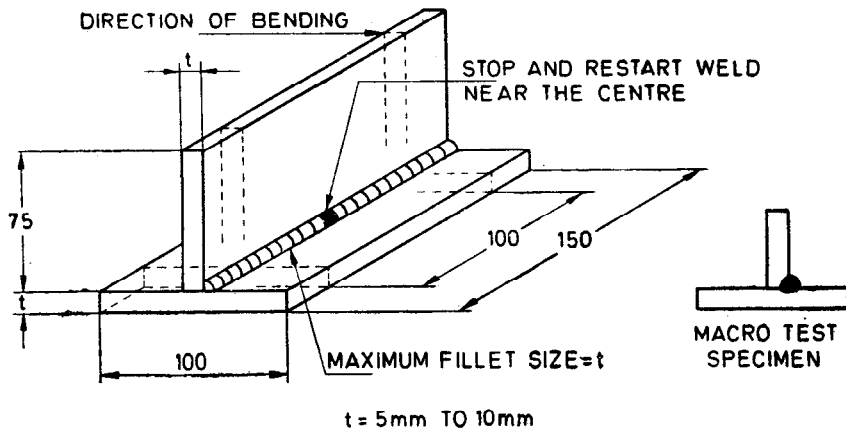
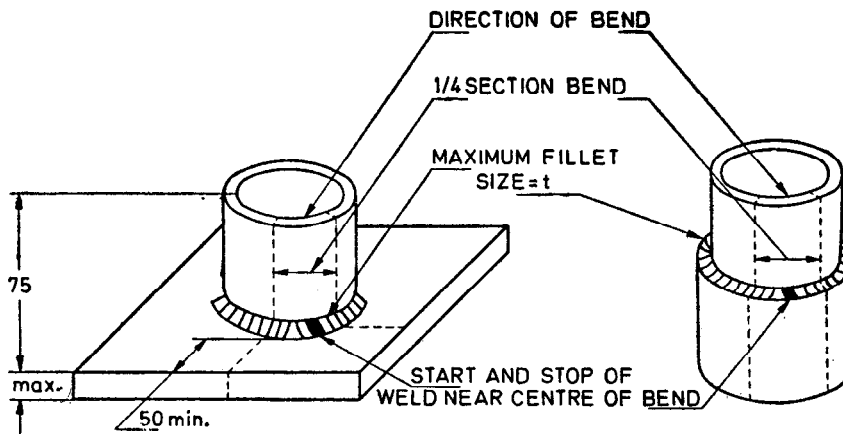


FIG. 7.12 FILLET WELD IN PLATE-PERFORMANCE QUALIFICATION



NOTE — Either pipe to plate or pipe to pipe may be used as shown.  
All dimensions in millimetres.

FIG. 7.13 FILLET WELD IN PIPE-PERFORMANCE QUALIFICATION

**7.8.4.3** The cut end of one of the end section shall be smoothed and etched with a suitable etchant ( *see* Appendix K ) to give a clear definition of the weld metal and heat affected zone. The visual examination shall show complete fusion and free from cracks except that the linear indications at the root shall not exceed 0.8 mm. The weld shall not have a concavity or convexity greater than 1.5 mm and the difference in length of legs of fillet shall not exceed 3 mm.

**7.8.4.4** For fillet welds the basic position for plates and pipes are described in Fig. 7.14 and 7.15.

**7.8.4.5** The limitations and number of tests required for fillet weld performance shall be as shown in Table 7.3 for plate, Table 7.4 for fillet qualification by groove weld and Table 7.5 for small diameter pipe fillet weld.

### **7.8.5 Limits of Qualified Positions**

**7.8.5.1** Welders who pass the required tests in groove welds in test positions ( *see* Fig. 7.1 ) shown in Table 7.6 shall be qualified for the position of groove welds, including branch welds, shown in Table 7.6 and the fillet welds ( *see* Fig. 7.14 and Fig. 7.15 ) shown in Table 7.6. In addition, the welders who pass the required test for groove welds shall also be qualified to make fillet welds and in all thicknesses and pipe diameters of any size within the limits of the welding variables ( *see* 7.8.7 ).

**7.8.5.2** Welders who pass required test for fillet welds in the test position of Table 7.6 shall be qualified for the positions shown in this table. Welders who pass the test for fillet weld on plate shall be qualified to make fillet weld only in all thicknesses of material, sizes of fillet weld and pipes and tubes 73 mm outside diameter and over, within the essential variable. Welders who make fillet weld in pipe or tube less than 73 mm outside diameter shall pass the pipe fillet test in accordance with 7.8.4.5.

**7.8.5.3** A fabricator who does production welding in a special orientation may make the test for performance qualification in this specific orientation. Such qualification is valid only for the position actually tested except that an angular deviation of  $\pm 15^\circ$  is permitted in the inclination of the weld.

### **7.8.6 Qualification Test Coupons**

**7.8.6.1** The test coupons may be plate or pipe. When all position qualifications for pipe are accomplished by welding one pipe assembly in both horizontal and multiple positions, large diameter pipes shall be employed to make up the test coupons as illustrated in Fig. 7.16.

**7.8.6.2** The dimensions of the welding groove of the test coupons used in making double welded groove welds or single welded groove welds with backing shall be the same as those of

any welding procedure or shall be as shown in Fig. 7.17. Partial penetration groove welders or fillet welds are considered as welding with backing.

**7.8.6.3** The dimensions of the welding groove of the test coupon used in making qualification tests for single welded groove welds without backing shall be the same as those for any WPS qualified by the manufacturer or as shown in Fig. 7.18.

### **7.8.7 Welding Variables**

**7.8.7.1** A welder or welding operator shall be qualified whenever a change is made in one or more of the essential variables listed. When a combination of welding process is required to make a weldment, each welder shall be qualified for the particular welding process or processes he will be required to use in production welding. A welder may be qualified by making tests with each individual welding process or with the combination welding process in a single test coupon. The limits of thickness for which the welder is qualified is dependent upon the thickness of test coupon as given in Table 7.6. The following are the essential variables and require requalification of welder or welding operator.

**7.8.7.2 Joints** — The deletion of backing and change in root detail in single welded groove welding. Double welded groove welds are considered as welding with backing.

**7.8.7.3 Base metal** — The change in the pipe diameter beyond the range qualified specified in Table 7.7 and thickness beyond the range specified in Table 7.2.

**7.8.7.4 Filler metal** — A change in classification of filler metal and a change in deposited weld metal thickness beyond the limit specified in Table 7.2. A change in more than one-third in the diameter of electrode for root run of manual metal arc welding.

**7.8.7.5 Positions** — The addition of other welding positions than those already qualified specified in Table 7.6 ( *see* 7.2, 7.8.4 and 7.8.7.7 ). For process other than submerged arc welding, a change from upwards to downwards or *vice versa* in the progression specified for any pass of a weld except that cover or wash pass may be up or down. The root pass may also be run either up or down when the root pass is removed to sound weld metal in the preparation for welding the second side.

**7.8.7.6 Shielding gas** — The emission of inert gas shielding ( backing ) except that requalification is not required when a qualified WPS is changed to emit the inert gas backing. This procedure is used for a single welded butt joint with a backing strip or a double welded butt joint or a fillet weld and in case of non-ferrous metals a change of more than 15 percent in the gas flow rate.

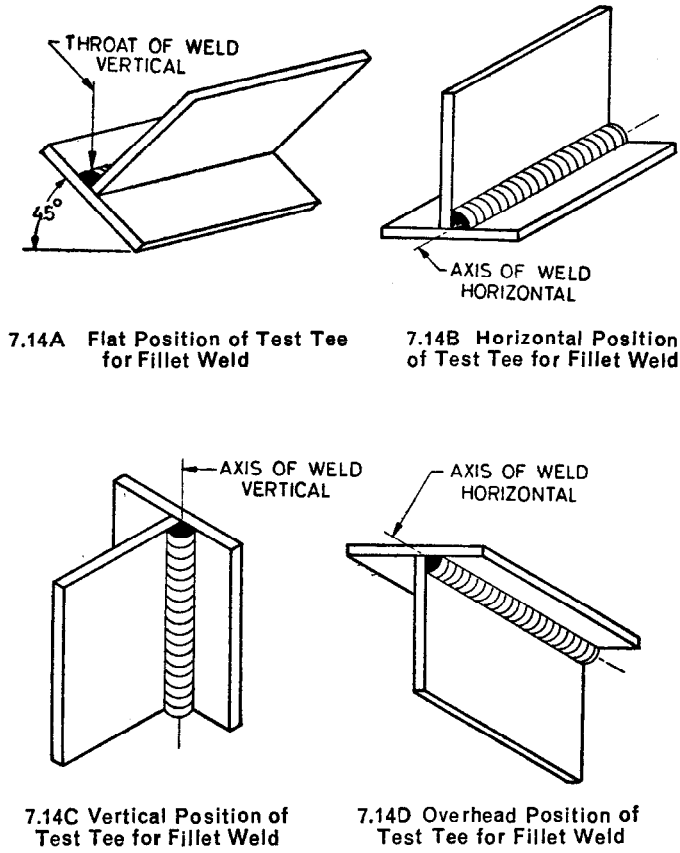


FIG. 7.14 POSITIONS OF TEST PLATES FOR WELDER PERFORMANCE QUALIFICATION OF FILLET WELDS

TABLE 7.3 PLATE FILLET-WELD TEST

| TYPE OF JOINT | THICKNESS <i>t</i> OF TEST COUPON AS WELDED<br>mm | RANGE QUALIFIED                                                                 | NUMBER OF SPECIMENS REQUIRED ( see FIG. 7.12 ) |          |
|---------------|---------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------|----------|
|               |                                                   |                                                                                 | Macro                                          | Fracture |
| Tee fillet    | 5 to 10                                           | All base material thicknesses, fillet sizes, and diameters 73 mm O. D. and over | 1                                              | 1        |

TABLE 7.4 FILLET QUALIFICATION BY PLATE OR PIPE GROOVE-WELD TESTS

| TYPE OF JOINT | THICKNESS <i>t</i> OF TEST COUPON AS WELDED | RANGE QUALIFIED                                            | TYPE AND NUMBER OF TESTS REQUIRED                                                                         |
|---------------|---------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Any groove    | All thicknesses                             | All base material thicknesses, fillet sizes, and diameters | Fillet welds are qualified when groove weld is qualified in accordance with either Table 7.1 or Table 7.7 |

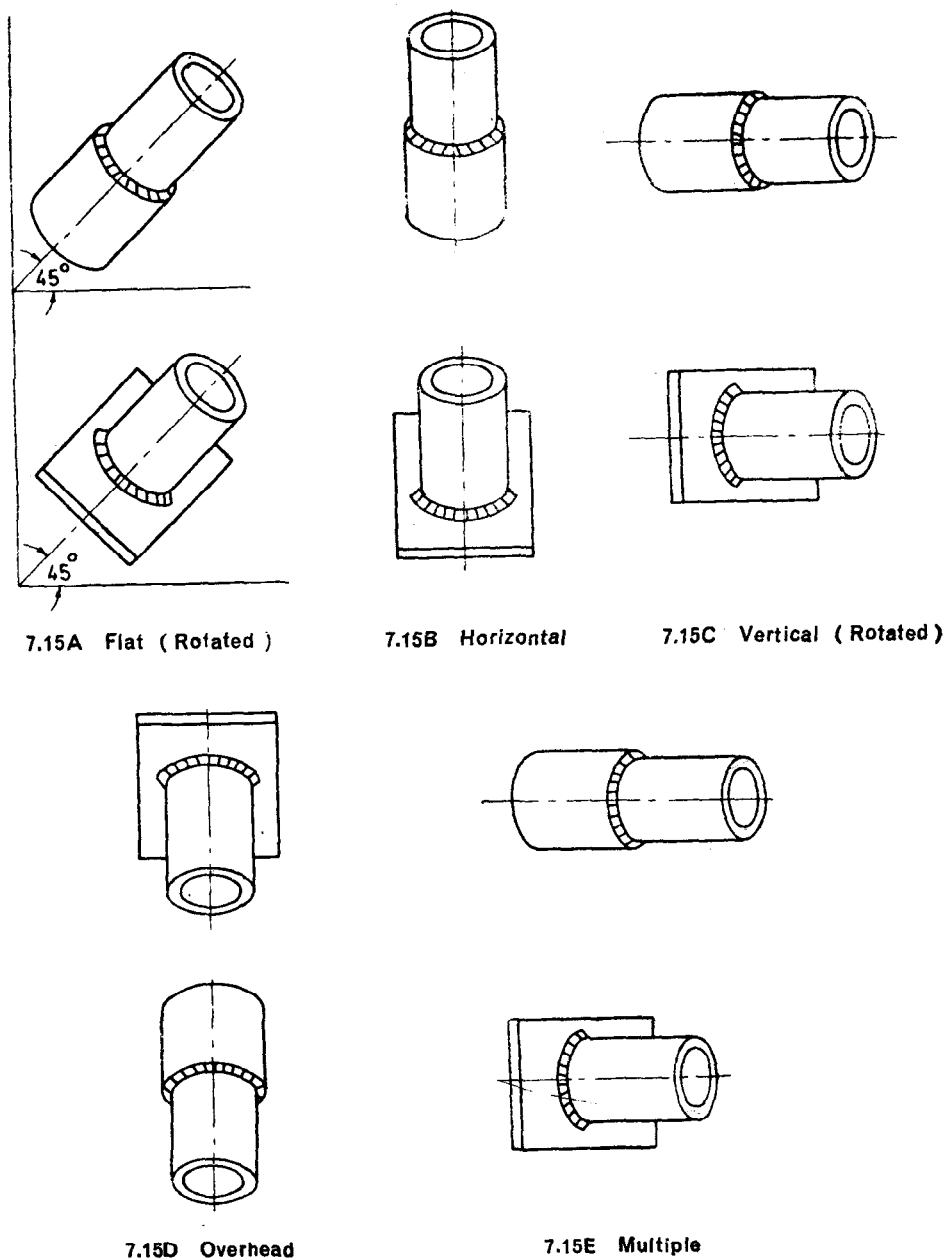


FIG. 7.15 POSITIONS OF TEST PIPES FOR WELDER PERFORMANCE QUALIFICATION OF FILLET WELDS

TABLE 7.5 SMALL DIAMETER PIPE FILLET-WELD TEST

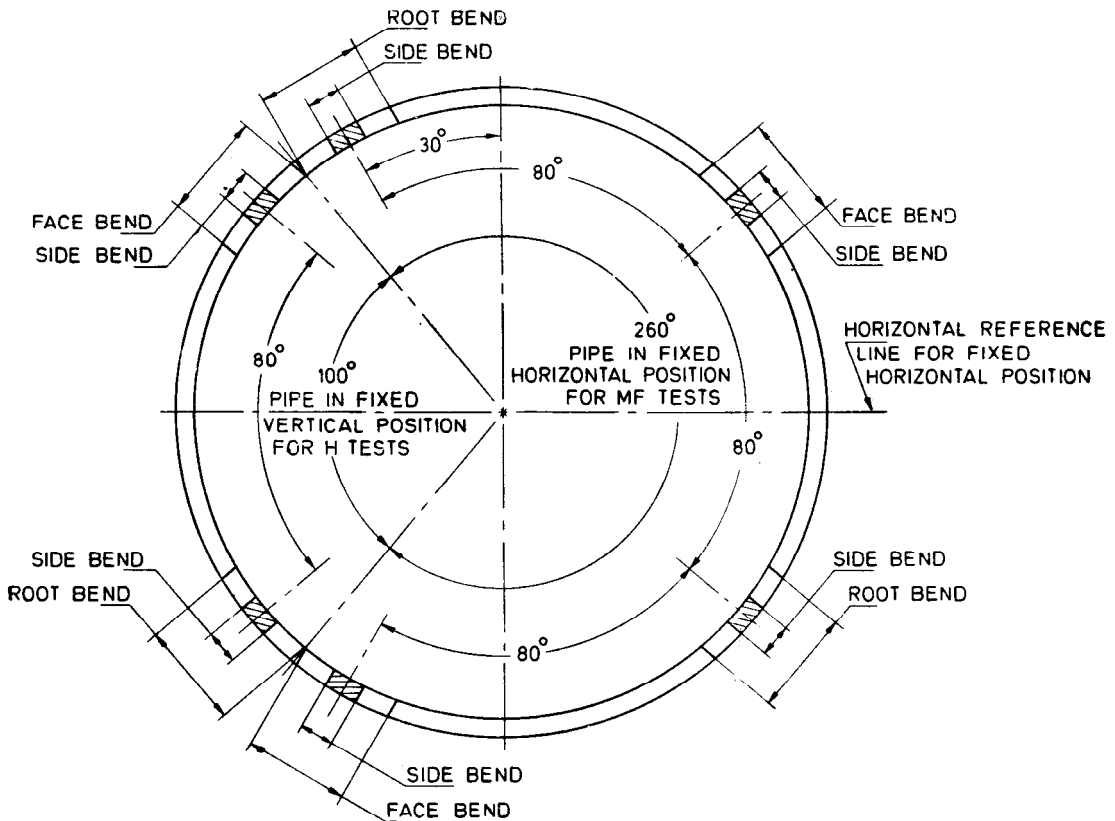
| NOMINAL PIPE SIZE OF<br>SAMPLE WELD | OUTSIDE DIAMETER QUALIFIED<br>(No Maximum) | PIPE WALL<br>THICKNESS<br>QUALIFIED | NUMBER OF SPECIMENS<br>REQUIRED (see Fig. 7.12) |          |
|-------------------------------------|--------------------------------------------|-------------------------------------|-------------------------------------------------|----------|
|                                     |                                            |                                     | Macro                                           | Fracture |
| Less than 20 mm                     | Minimum of not less than<br>size welded    | All                                 | 1                                               | 1        |
| 20 to 50 mm                         | Over 25 mm                                 | All                                 | 1                                               | 1        |
| Over 50 mm                          | 73 mm and over                             | All                                 | 1                                               | 1        |

**TABLE 7.6 PERFORMANCE QUALIFICATION POSITION LIMITATIONS**

| QUALIFICATION TEST |                   | POSITION AND TYPE WELD QUALIFIED ( NOTE 1 ) |                        |                           |
|--------------------|-------------------|---------------------------------------------|------------------------|---------------------------|
| Weld               | Position          | Groove<br>Plate and Pipe<br>Over 600 mm OD  | Pipe                   | Fillet<br>Plate and Pipe  |
| Plate-Groove       | <i>F</i>          | <i>F</i>                                    | <i>F</i> ( Note 2 )    | <i>F</i>                  |
|                    | <i>H</i>          | <i>F, H</i>                                 | <i>F, H</i> ( Note 2 ) | <i>F, H</i>               |
|                    | <i>V</i>          | <i>F, V</i>                                 | <i>F</i> ( Note 2 )    | <i>F, H, V</i>            |
|                    | <i>O</i>          | <i>F, O</i>                                 | <i>F</i> ( Note 2 )    | <i>F, H, O</i>            |
|                    | <i>V and O</i>    | <i>F, V, O</i>                              | <i>F</i> ( Note 2 )    | All                       |
| Plate-Fillet       | <i>H, V and O</i> | All                                         | <i>F, H</i> ( Note 2 ) | All                       |
|                    | <i>F</i>          | —                                           | —                      | <i>F</i> ( Note 2 )       |
|                    | <i>H</i>          | —                                           | —                      | <i>F, H</i> ( Note 2 )    |
|                    | <i>V</i>          | —                                           | —                      | <i>F, H, V</i> ( Note 2 ) |
|                    | <i>O</i>          | —                                           | —                      | <i>F, H, O</i> ( Note 2 ) |
| Pipe-Groove        | <i>V and O</i>    | —                                           | —                      | All ( Note 2 )            |
|                    | <i>F</i>          | <i>F</i>                                    | <i>F</i>               | <i>F</i>                  |
|                    | <i>H</i>          | <i>F, H</i>                                 | <i>F, H</i>            | <i>F, H</i>               |
|                    | <i>MF</i>         | <i>F, V, O</i>                              | <i>F, V, O</i>         | All                       |
|                    | <i>MI</i>         | All                                         | All                    | All                       |
| Pipe-Fillet        | <i>H and MF</i>   | All                                         | All                    | All                       |
|                    | <i>F</i>          | —                                           | —                      | <i>F</i>                  |
|                    | <i>H</i>          | —                                           | —                      | <i>F, H</i>               |
|                    | <i>V</i>          | —                                           | —                      | <i>F, H</i>               |
|                    | <i>O</i>          | —                                           | —                      | <i>F, H, O</i>            |
|                    | <i>V and O</i>    | —                                           | —                      | All                       |

NOTE 1 — Position of welding shown in Fig. 7.1 and Fig. 7.14.

NOTE 2 — Pipes 73 mm OD and over.

*F* = Flat*H* = Horizontal*V* = Vertical*O* = Overhead*MF* = Multiple Fixed*MI* = Multiple Inclined

NOTE 1 — Side bend may be substituted for face and root bends for thickness above 10 mm.

NOTE 2 — Acceptance as per 7.6.

**FIG. 7.16 ORDER OF REMOVAL OF TEST SPECIMENS FROM PIPE — ALL POSITION-PERFORMANCE TEST**

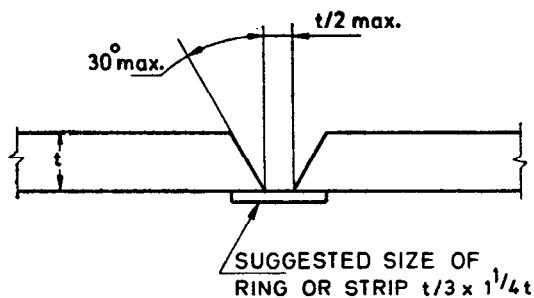


FIG. 7.17 SINGLE WELDED GROOVE-JOINT WITH BACKING

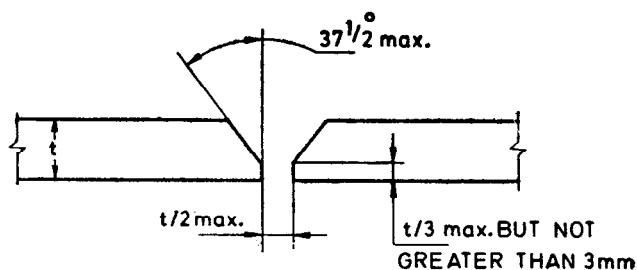


FIG. 7.18 SINGLE WELDED GROOVE-JOINT WITHOUT BACKING

TABLE 7.7 PIPE GROOVE — WELDS DIAMETER LIMITS — PERFORMANCE QUALIFICATION

| NOMINAL PIPE SIZE OF SAMPLE WELD | OUTSIDE DIAMETER QUALIFIED (NO MAXIMUM) mm |
|----------------------------------|--------------------------------------------|
| Less than 20 mm                  | Minimum of not less than size welded       |
| 20 to 50 mm                      | Over 25 mm                                 |
| Over 50 mm                       | 73 mm and over                             |

**7.8.7.7 Electrical characteristics** — In case of gas shielded welding a change from ac to dc or *vice versa* and in dc welding a change in the electrical polarity. For gas shielded arc welding a change from spray arc, globular arc or pulsating arc to short circuiting arc or *vice versa*.

**7.8.7.8 Non-essential Variables** — Variables other than above are non-essential variables including changes of location and does not require fresh performance qualification for metal arc welding, submerged arc welding and inert gas arc welding. For other welding processes this is to be drawn in consultation with the Inspection Authority.

**7.8.8 Re-tests** — A welder or welding operator who fails to meet the requirement of one or more of the test specimens prescribed in Table 7.2 may be retested under following conditions.

**7.8.8.1** When the qualification coupon has failed in mechanical testing in accordance with 7.8.4.1, retesting shall be by mechanical testing. When an immediate retest is made, the welder or welding operator shall make two consecutive test coupons for each position for which he has failed, all of which shall pass the test requirement.

**7.8.9 Renewal of Qualification** — The performance of the welder or welding operator shall be affected under the following conditions.

**7.8.9.1** When he has not welded with a process during a period of 3 months or more, his qualification for that process shall be expired, except when he is welding with another process, the period may be extended to 6 months.

**7.8.9.2** When he has not welded with any process during a period of 3 months all his qualification shall be expired including any of which may have extended beyond 3 months by virtue of 7.8.9.1.

**7.8.9.3** When there is a specific reason to question his ability to make welds that meet the specification which support the welding that he is doing his qualification be considered expired.

**7.8.9.4** Renewal of qualification for a specific welding process under 7.8.9.1 or 7.8.9.2 may be made in a single test joint, plate or pipe on any thickness, position or material to re-establish welder or welding operator qualification for any thickness, position or material for the process for which he was previously qualified.'